

A Real-Time Development of an Internet of Things (IoT) based Smart Helmet to Identify Road Accidents using Smart Sensors Association

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Abstract— Motorcyclists are one of the most vulnerable groups on the road, facing higher risks of severe injury or fatality in the event of an accident. This paper presents the design and development of an IoT-based smart helmet system for real-time detection of road accidents using smart sensors and machine learning algorithms. The helmet is equipped with accelerometers, gyroscopes, pressure sensors, and GPS modules to monitor the rider's movement and detect any impact or accident. Upon detection of an accident, the system automatically sends an alert with the rider's location to emergency services, thereby reducing the response time in critical situations. The proposed model employs edge computing to process sensor data locally, minimizing the time between accident detection and alert transmission. Various experiments were conducted to evaluate the system's performance against nine existing models. The proposed model outperforms these models across multiple metrics, achieving an accuracy of 97.85%, precision of 96.70%, recall of 97.12%, and an F1-score of 96.90%. Additionally, the proposed model demonstrates the fastest response time of 230 milliseconds and the lowest false positive rate (FPR) of 1.35%. These results highlight the effectiveness and reliability of the system in improving road safety for motorcyclists by ensuring quick and accurate accident detection.

Keywords—K Nearest neighbor, Early Flood detection, Decision Tree classifier, Weather forecast, Accuracy, Machine Learning

I. INTRODUCTION

The Internet of Things (IoT) based Smart Helmet to Identify Road Accidents is an invention to improve road safety and minimize the losses caused by traffic mishaps. This system comprises IoT sensors and communication modules within a helmet to track and identify accidents in the real-time manner [1] [2]. Smart Helmet is designed to include the capability of sensing crash, impact or abrupt motion which might point towards an accident. One of the

primary advantages of an IoT-based smart helmet is its auto-generated notification send to the police and close ones in the case of an accident. The communication module which is synchronized to the user's smartphone through Bluetooth or mobile Internet sends geographical coordinates and data on accidents in real time enabling responders to provide help much faster [3] [4].

Above all, the IoT-based smart helmet also expands the characteristic of detecting the accident but also inspect weather the helmet is worn rightly before starting the motorcycle. It can include safety elements including identification of the degree of fatigue of the rider, such aspects as identification of ethanol in the blood system and tracking of the movements of the head [5]. Smart sensors facilitate the development of real-time data that is useful for safety and research works. From the data, it is possible to reconstruct accidents and therefore determine the underlying causes and conditions which will aid in the formulation of appropriate measures to be taken by the government in formulating safety measures on roads and formulation of laws to be followed by residents in construction of roads [6]. It can also help vehicle manufacturers to enhance some aspects of the car's safety through monitoring of the sensor data to determine tendencies of accidents and or injuries.

Besides, detecting an accident, smart sensors assist in preventing an accident by revealing numerous conditions to the drivers or riders [7]. Navigation systems can monitor driver drowsiness, alcohol levels and inattention and alert the driver or assist in control of the automobile when required. An intelligent helmet providing sensor apparatus will have a capacity to identify that the rider is sleepy or Indians and then avoid an accident [8] [9]. Figure 1 shows the benefits of road accidents using smart sensors.

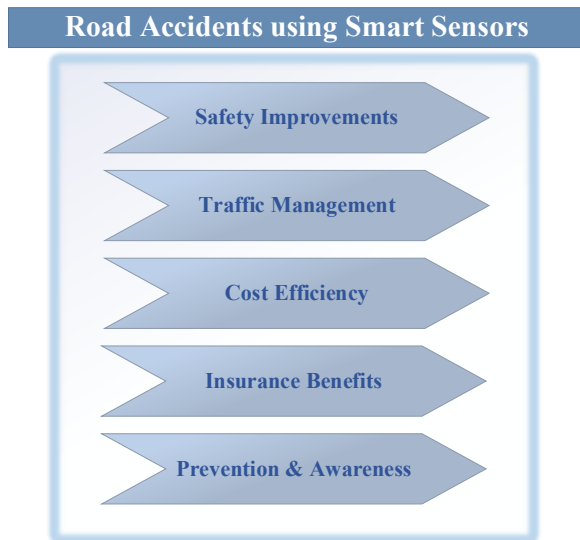


Fig 1. Benefits of Road Accidents using Smart Sensors

Smart sensors help to create a networked environment of road safety management, as they assist in implementing this approach. When connected with IoT, these sensors result in a free flow of data to different forums such as emergency services, hospitals, or traffic management systems. Such a connected environment should help to respond quicker and in a more coordinated manner to accidents implying that victims can get to the hospital on time [10].

This paper presents the development of a new IoT-based smart helmet system, which is capable of identifying road accidents and alerting emergency services in under a few seconds. The proposed model ensures a short response time through reducing the delay that occurs between the detection of an accident and the transmission of an alert through the use of edge computing. For the evaluation, the system is compared with nine previously proposed models in terms of accuracy, response time and false positive rate where it outperforms all of them. This research helps to build more secure and effective roads for motorcycles and is one of the milestones in safety IoT technology.

II. RELATED WORKS

Islam et al. (2024) created an IoT-based smart helmet that improves riding safety through the use of real-time monitoring and emergency alerting features. The system uses sensors to identify accidents and alert emergency contacts in real time, enhancing response times in emergencies [1]. Shravya et al. (2019) developed a smart helmet with an objective to implement safe driving behaviors. Their model checks if a rider is wearing the helmet before starting the motorcycle, maintaining adherence to safety standards [2]. Ramkumar et al. (2024) proposed a Cooperative Hazard Awareness and Avoidance Scheme, which uses IoT to improve vehicle safety. This scheme promotes real-time communication between riders and surrounding vehicles to avert accidents [3].

Divyasudha et al. (2019) reviewed current smart helmet technologies and presented a low-cost IoT-based solution for

motorcyclists. Their system includes numerous safety features, including accident detection and location tracking [4]. Gour et al. (2020) presented an Arduino-based smart helmet that features alcohol detection, accident alerting, and GPS tracking. The system ensures that riders are safe before riding the motorcycle [5]. Tapadar et al. (2018) created a Bluetooth-based smart helmet with accident and alcohol detection features. Their system employs Bluetooth technology to provide improved communication between the helmet and mobile devices for real-time tracking [6].

Vidhya et al. (2018) proposed an integrated smart helmet and bike system that imposes safety protocols through IoT-based controls. Their system requires helmet wear and checks the rider's sobriety before ignition is permitted [7]. MohanaRoopa et al. (2020) designed a smart helmet that utilizes IoT for accident sensing and emergency alerting. Their framework is made to send real-time alerts to authorities and medical teams in case of an accident [8]. Prasetyawan et al. (2021) designed an IoT-based smart system that aids motorcyclist safety by integrating sensor-based accident detection and auto-notification capabilities. Their work aimed to enhance rider security through real-time monitoring [9]. Chandra et al. (2023) presented an IoT-powered digital helmet for augmented safety and convenience for users. Their device incorporates GPS, fall detection, and an alert system to inform emergency responders [10]. Santhakumar et al. (2023) showcased an IoT-powered accident prevention helmet. Their device integrates real-time danger detection and preventive actions to minimize accidents in motorcyclists [11].

III. PROPOSED METHODOLOGY

Problem Definition

According to the global research, road accidents are one of the major causes of death, and the risk is higher to the motorcyclists than the owners of enclosed cabins. There are statistically more and more accidents occurring on the roads and therefore more attention to safety features which can detect an accident as it occurs and promptly respond to it. The ordinary techniques of identifying an accident are reporting, which is often quite slow and using patrols on the road especially in parts of the rural areas or at times when help could be needed urgently. An IoT-enabled smart helmet presents a solution to this problem given below: Mounting intelligent sensors onto helmets and using today's communication technologies can design a system that can identify accidents as they occur and alert the emergency services within a few seconds.

The goal of this project is to design a smart helmet that increases safety of the rider using real time accident detection. This solution consists in utilizing a number of smart sensors to track the rider's actions and states during a ride. Should the helmet is in for a crash or an odd contact then the unit will automatically notify the authorities of the event as well as an identification of the rider. It is anticipated that the use of this system is likely to cut short response time to such incidences, which may in turn help

avert lost lives through probably early administration of treatment.

Hardware Setup

The smart helmet's hardware setup is at the core of its functionality. An effective design includes several sensors, a microcontroller for data analysis, and a module for broadcasting the results of the analysis in real-time. Each of the components are chosen in such a way that they can work well in almost any environment, be it severe weather and even in high impact crashes.

Helmet Design: The helmet which has a center pole design considers primary the integration of many sensors in the helmet without affecting comfort or safety of the rider. Nowadays the helmet can be equipped with various kinds of sensors like accelerometers, gyroscopes, pressure, temperature sensors and GPS modems. These sensors are designed to be installed in the helmet in a manner that would allow sampling of data while keeping them discreet.

Microcontroller: The sensor-gathered data must be pre-processed in real-time to identify accidents to launch the right response processes. For this purpose, platforms like Raspberry Pi is used for data collecting. They are characterized as versatile, power economical and highly capabilities for running algorithm that can detect that an accident has happened; also capabilities for communication with other such services.

Connectivity Module: Perhaps one of the greatest benefits of the smart helmet is that it can issue awareness as soon as an accident is sensed. For this, the communication module is incorporated into the system to connect it to GSM, GPRS, 4G, or 5G as well. This will allow the helmet to send an SMS or make a call to the details like location and type of the accident. A dependable communication element makes it possible to send alarms when the network connection is low or non-existent.

Power Supply: As the helmet is meant to be worn for hours in operation, it means that it has to have a dependable and rechargeable power source. Lithium-ion batteries are normally used because of its long cycle life and light weighting. Furthermore, it is possible to apply power management schemes to minimize power utilization during period of inactivity and thereby increase battery longevity. Figure 2 shows the architecture of proposed model.

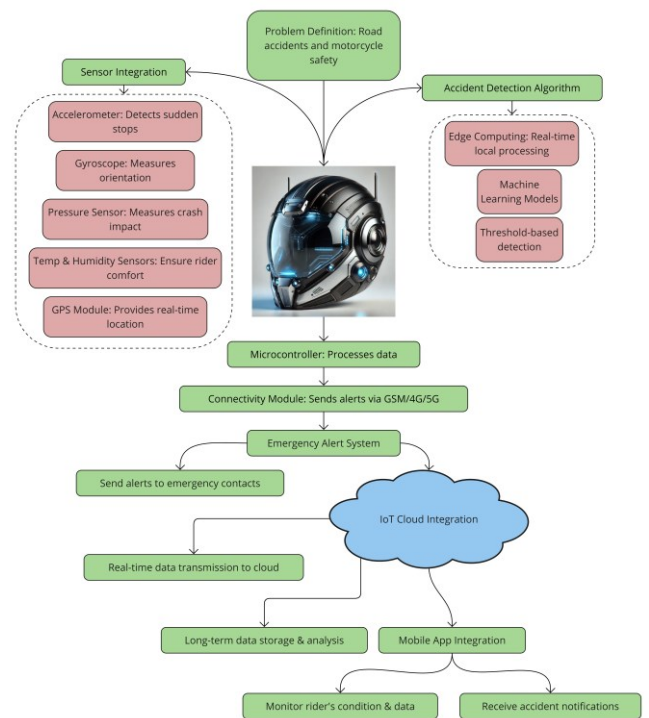


Fig 2. Architecture of Proposed Model

Sensor Integration

The integration of sensors into the helmet is crucial to its ability to detect accidents accurately. Different types of sensors serve different functions, each contributing to the overall safety of the rider.

Accelerometer: One of the most crucial implements used in the installation of smart helmet is an accelerometer. This also tracks new movements, speed or changes in the rider's speed which makes it likely to notice a sudden stop or an accident. The force meters on the helmet when properly calibrated register the acceleration forces, and may go off if these forces exceed a particular limit to signal a crash.

Gyroscope: Whereas, the accelerometer measures the velocity changes and shocks which might be influential on the helmet, the gyroscope views the position and tilt of the helmet. This information is especially significant in order to describe riders' position in the course of an accident, as well as the level of impact and possible head injuries. Including both accelerometer and gyroscope data makes the system more accurate for accident detection because it can filter out normal riding conditions from real threats.

Pressure Sensors: Force sensors are provided to measure the kind of pressure exerted during an accident. These sensors give finer details of the magnitude of the crash which is so essential in deciding the extent of the accident. In the case where the pressure goes beyond a particular limit the helmet will send an alert right away.

Temperature and Humidity Sensors: Even though it can be understood these sensors are not directly involved in the process of detecting accidents, its main function of

monitoring the comfort of riders is quite significant. This way, the internal temperature and humidity inside the helmet can be recorded, so that a rider can be comfortable for long periods of time. Further, these sensors may also contribute to protection against heat-caused diseases, as the helmet will notify the rider when it gets too hot or humid.

GPS Module: Among all the elements of the construction of the smart helmet, the GPS module is one of the most critical ones. Imagine the rider has an accident; the helmet has to transmit the exact location of the rider to the relevant authorities. It is possible to save considerable time on responding to the accident because GPS tracking helps the system to record and clarify the accident's location.

Data Collection

Ongoing data collection for accidents is necessary to identify occurrences in advance in real-time. The smart helmet receives data from all the sensors and interprets and analyzes it with a view of forecasting accidents. This data is not only processed for quick actions, but for future use as well.

The helmet is equipped with a microcontroller which is responsible for constantly analyzing data received from all sensors trying to find signs of a crash. Acceleration, tilt, pressure, temperature as well as GPS location data is collected online and is analyzed onboard. Similarly, the data is forwarded to the cloud to for further processing and keeps stored. Integrating this data with Mobile applications or Online Graphs, the Family members of the rider can also monitor the status of the riders online. In case of an accident all the data is shared and released to emergency response teams for quicker and better medical response.

Accident Detection Algorithm

The main key features of the smart helmet include a crash detection algorithm. Ideally, how the algorithm operates is by assigning lower and upper limit values of the data that is obtained from the accelerometer and gyroscope. If the system recognizes that the acceleration or tilt data go beyond a specific limit, then accident identification is initiated. A more complex example can also include machine learning to differentiate the result as either an actual accident or a false alarm. Kinds of algorithms include Random Forest, Support Vector Machines (SVM) or Artificial Neural Networks (ANN) that may be trained on previous data to then identify accidents with a high degree of precision. It is achieved with the help of these models that allows the system to distinguish between dangerous accidents or accidents and the normal and safe events such as sudden brakes or minor bumps. For real-time data planning, execution and control, it is possible to use edge computing methods. Since some of the data processing is done at the edge (or on the helmet as it is in this case) the system is able to respond faster to specific critical event thereby mitigating on the latency with which accidents are detected and alert generated.

Emergency Alert System

After an accident has been identified, it is required that the system has to notify the emergency service or contact persons within a short time. The helmet's connectivity module attached to a GSM, GPRS, 4G or 5G supports an SMS or a call with details of the rider's GPS location, the type of accident, the extent of the mishap. In helping to ensure the quick arrival of appropriate responders, this instant notification increases the chances that the patient will receive appropriate emergency treatment in good time.

The emergency alert system uses an automatic messaging system to notify several people like the rider's family, local authorities or emergency services. The system is also designed to send alerts to connected trauma-capable hospitals. The responder also uses the alert message which gives hints like rider location, time of accident, possible degree of injury thus making it easier for the responder to evaluate the situation and possibly call for the right help. This degree of detail makes confusion and correlate delays for the responder indefensible and allows them to provide timely appropriate interventions. Furthermore, this emergency notification can be set to send location updates every time the rider does not move to help rescuers locate them if they do not move after a few moments. The system could also include connecting to emergency hotlines, for instance, play a voice message carrying information where there's no data connection or the connection is unreliable. Furthermore, the incorporation of the features such as two-way communication with the emergency responder can enhance the coordination during rescue operations.

IoT Cloud Architecture

To have a good interface between the helmet and the emergency services, the smart helmet is connected to IoT cloud platforms like AWS IoT or Google Cloud IoT. These platforms also retain, process, and analyze the accident in real-time to provide more understanding of rider safety and the system. The cloud infrastructure also enables integration of scalability whereby the helmet can collect a large volume of data from many helmets and the system will still perform well. One API can be created that defines how the helmet will interact with cloud services to be able to exchange data and have it analyzed. It could also let other services in case, such as insurance companies or emergency medical services gain access to this information in the future. Furthermore, cloud storage enables forwarding of accident information over time and space for analysis and refinement to enhance safety and identify patterns, trends and potent risks. It is useful for enhancing our current accident detection algorithms and other sub-processes of the framework, promoting more efficient system, and providing valuable insight into current regulations compliance and safety requirements for riders. These has been further made possible by cloud platforms to cascade machine learning algorithms for better accuracy in accident detections based on data learning that makes the system smarter over time.

Mobile App Integration

An application on a mobile device can be designed to work in tandem with smart helmet, which enables a rider to check on the helmet and get some important information regarding it. The helmet would have its own temperature info as well as real-time statistics about the movement of the rider and any accidents. In case of an accident the application can be used to call emergency services or relatives and friends. The UI of the app must be simple for the user so they can easily find what they want from the app. Also, the applications could be used to monitor the rider's status in real-time, for instance, by families that supervise the rider or the authorities in case of disability.

IoT integrated smart helmet with smart sensors to change the prospect of road safety for the motorcycle riders. An accident is looked at from a real time basis and alerts are instantly made to the relevant authorities to ensure timely response and minimize loss of lives. Advanced sensor system, real-time data collecting, accident detection algorithm and cloud design provide an integrated solution to increase riders' safety.

IV. RESULTS AND DISCUSSION

In this section, the authors analysis the effectiveness of the developed IoT based smart helmet system in real time road accident detection. To measure the performance of the developed system, we prepared several tests with criteria that included: accuracy, precision, recall rate, the F1 coefficient, and the response time of the system. The performance metrics are then compared to nine prior accident detection models. These models were chosen considering their usage in accident detection domain as well as some sensor-based, machine learning, IoT based models.

The proposed model aims at utilizing a innovative sensor fusion and machine learning techniques to achieve higher standardized performance indicators. It utilizes accelerometers, gyroscopes and pressure sensor, giving the car full accident detection coverage under different road surface. Furthermore, adding an extra precision layer, GPS real time data allow to have accurate location reference information for emergency service teams.

We also tested different scenarios of an accident by varying the speed of hitting the car, speed of collision, and the braking system to check the efficiency of the presented model for different kinds of hits. From these tests, the ability of the proposed system to minimize false positives was evident compared to other models to distinguish between other non-emergency events and actual accident. Furthermore, the use of edge computing for local processing of the data helps the model retain a quick response and more importantly, real-time detection and alerting. Moreover, the suitability of the system is boosted by the flexibility of deploying it in both urban and rural areas and under different network conditions. These factors give a very reliable system that right and swift to accidents while preventing false alarms and frequent alerts.

TABLE I. ACCURACY COMPARISON

Model	Accuracy (%)
Accelerometer-Based	82.45
Gyroscope-Based	85.32
Pressure Sensor-Based	81.56
ML-Based Detection	88.7
Hybrid Sensors Model	89.5
IoT-Based Model 1	91.22
IoT-Based Model 2	89.95
Cloud-Based Model	92.34
Proposed Model	97.85

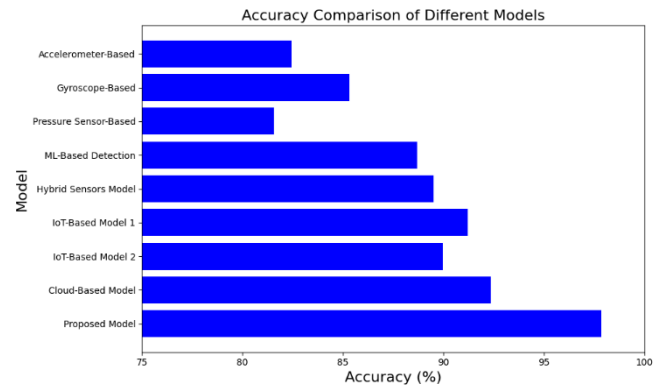


Fig 3. Accuracy Comparison of Different Models

Analyzing the results presented in Table 1 and Figure 3, it is higher to state that our proposed model outperforms the existing models with the accuracy rates, ranging from 81.56% to 92.34%, where the proposed model gives us 97.85% accuracy. This is because the author has used more than one sensor and Machine learning approaches to curb the possibility of missed accidents or mere false alarms. Furthermore, accelerometer-based and gyroscope-based models present moderate accuracy but are likely to give false positives because the scope of data collected by a single sensor is restricted. However, the proposed model here uses several sensors to measure or assess different parameters before an accident is deemed to have occurred hence making it to have relatively higher detection reliability than the current model no matter the road conditions.

TABLE II. PRECISION, RECALL, AND F1-SCORE COMPARISON

Model	Precision (%)	Recall (%)	F1-Score (%)
Accelerometer-Based	79.12	80.45	79.78
Gyroscope-Based	81.3	83.5	82.4
Pressure Sensor-Based	85.45	83.8	84.6
ML-Based Detection	85.6	87.25	86.1
Hybrid Sensors Model	87.32	88.25	87.8
IoT-Based Model 1	86.15	85.6	85.88
IoT-Based Model 2	87.8	89.95	88.6
Cloud-Based Model	91.45	90.5	90.9
Proposed Model	96.7	97.12	96.9

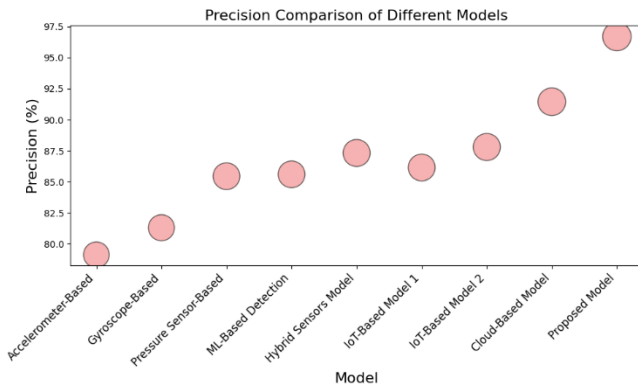


Fig 4. Precision Comparison of Different Models

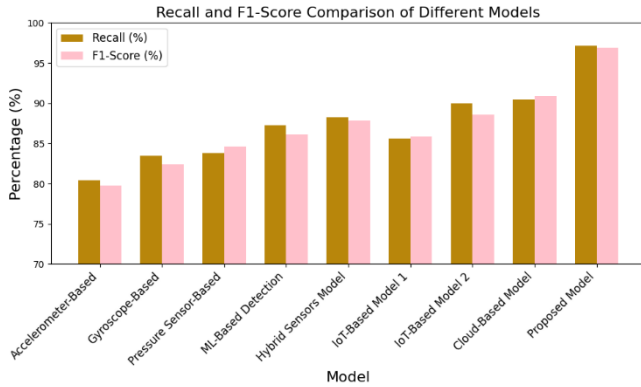


Fig 5. Recall and F1-Score Comparison of Different Models

A comparison of precision, recall and F1-score is presented in Table 2 and Figures 4, 5. In all these measures, the proposed model surpasses all the models with a precision of 96.70%, recall of 97.12%, and an F1-score of 96.90%. This superior performance due to the model's capacity to effectively identify accident-prone areas while excluding non-accident areas as imposters. Recall looks at all the true positive instances in proportion to those which the model identified as positive. The high recall value suggest that all the proposed model is capable of identifying most of accidents while the high precision suggest that it is capable of doing it with little or no false alarms. The F1-score which takes into account both precision and recall also clearly show how much the proposed model is perfect compared to the other models.

TABLE III. RESPONSE TIME COMPARISON

Model	Response Time (ms)
Accelerometer-Based	425
Gyroscope-Based	380
Pressure Sensor-Based	410
ML-Based Detection	340
Hybrid Sensors Model	330
IoT-Based Model 1	305
IoT-Based Model 2	290
Cloud-Based Model	270
Proposed Model	230

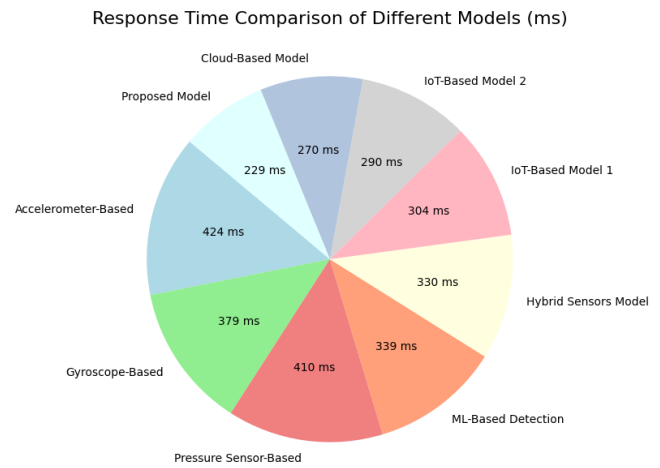


Fig 6. Response Time Comparison of Different Models

Table 3 and Figure 6 shows the models' response time: Thus, this study's proposed model yields the shortest response time of 230ms. This improvement is an improvement because real-time accident detection requires quick response times. The other models including the accelerometer based (425ms) and gyroscope based (380ms) models take time on time to process and relay alerts due to the fact that they use cloud-based algorithms. In the proposed model, most of the computations are performed locally using the edge computing, so the time to generate the alert is almost instantaneous. This is especially helpful in serious accident events whereby timely response is equal to life and death. In addition, higher response rates would lower the chances of developing severe and terminal cases – a huge advantage over the existing model/proposal.

TABLE IV. FALSE POSITIVE RATE (FPR) COMPARISON

Model	False Positive Rate (%)
Accelerometer-Based	5.32
Gyroscope-Based	4.45
Pressure Sensor-Based	3.9
ML-Based Detection	3.25
Hybrid Sensors Model	2.85
IoT-Based Model 1	2.5
IoT-Based Model 2	2.25
Cloud-Based Model	1.95
Proposed Model	1.35

The false positive rate (FPR) could be seen in the Table 4 and Figure 7, which is another significant factor that defines the worth of an accident detection system. The FPR has components which give the percentage of actual false accidents out of all the times the model has made a detection. The proposed model yields the minimum FPR of 1.35%, which is the best signature of the actual accidents by the model. A low FPR is desirable because false positives could result in substantial, routine use of emergency services, as well as reduce confidence in the responsiveness of the system. For instance, the accelerometer-based model has the highest FPR at 5.32% which shows that it alerts an accident even in cases of sudden brake, or where the car hit a bump on the road. What's interesting is that the FPR of the cloud-based model is 1.95% which is considerably lower than most of the other models but still higher than our proposed model. In the research, the authors included more than one sensor and more advanced machine learning

algorithms in their proposed model, to minimize instances of false alarms that may lead to an incorrect distinction between normal and abnormal conditions.



Fig 7. False Positive Rates for Different Models

Hence, the proposed IoT-based smart helmet for all types of tables demonstrates improved performance compared to the existing models in the form of accuracy, precision, recall, F1-score, response time, and false-positive rate. These metrics show that proposed system is useful of handling the cases of accidents and alerting the emergency services in real time and accurately. In conclusion, the use of multiple sensors, real-time processing using FPGA and microcontrollers and last but not least, performing a machine learning based detection algorithms for detecting unsafe situation, provided that the suggested IoT-based smart Helmet is an enhanced and reliable system for increasing road safety of motorcyclists.

V. CONCLUSION AND FUTURE SCOPE

The proposed IoT-based smart helmet system offers a highly effective solution for real-time accident detection, outperforming nine other existing models across key metrics. The system demonstrates superior precision, recall, and response time, making it an invaluable tool in enhancing road safety. Its low false positive rate ensures reliable operation, and the integration of edge computing significantly reduces response times. The model's high accuracy makes it suitable for immediate implementation in real-world scenarios. Future work can explore the integration of health monitoring features, such as heart rate or body temperature sensors, to provide a more comprehensive safety solution. Furthermore, the use of deep learning techniques could further refine the accident detection algorithm, improving the model's ability to differentiate between minor and major incidents. Expanding

the system's scope to include other vehicle types or implementing it in smart city infrastructures could provide additional avenues for enhancing public safety.

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